Here is the comprehensive documentation of the details and formulas based on the research article[1].

1. Solar PV Panel Technical Specifications

The calculations rely on the specific characteristics of the 10W panels used in the study[2].

Parameter	**Symbol**	**Value (at STC)**
Maximum Power	P_m	10 W
Maximum Power Voltage	V_m	20 V
Maximum Power Current	I_{im}	0.50 A
Short Circuit Current	I_{sc}	0.60 A
Open Circuit Voltage	V_{oc}	24.8 V
Temp. Coefficient (P_m)	β_p	$-0.3\%/^{\circ}C$ to $-0.5\%/^{\circ}C$
Temp. Coefficient (I_{im})	α_{Im}	$+0.1\%/^{\circ}C$
Temp. Coefficient (V_m)	β_{Vm}	$-0.38\%/^{\circ}C$

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2. Core Mathematical Formulas

A. Cell Temperature (T_{cell})

The real-world temperature of the solar cell is calculated based on ambient conditions and irradiation4.

$$T_{cell} = T_{\infty} + 0.078 H_{g}$$

- **T_{∞}**: Ambient air temperature ($^{\circ}C$)[5].
- **H_g**: Global inclined solar irradiance (W/m^2)[6].

B. Real Power Output (P_{pv})

This formula estimates power under real operation and climatic conditions7.

$$P_{pv} = P_{stc} \times DF_{pv} \times (H_g / H_{stc}) \times [1 + \beta_p(T_{cell} - T_{stc})]$$

- **P_{stc}**: Nominal PV power at STC (10 W)[8].
- **H_{stc}**: Standard solar radiation (1000 W/m^2)[9].
- **T_{stc}**: Standard test temperature (25 °C)[10].
- **DF_{pv}**: PV derating factor[11].

C. Output Current (I) - Single Diode Model

The standard electrical behavior of the PV cell is represented by this equation12.

$$I = I_{ph} - I_0 \left(e^{\frac{q(V + I R_s)}{n k T}} - 1 \right) - \left(\frac{V + I R_s}{R_{sh}} \right)$$

- **I_{ph}**: Photocurrent[13].
- **I_0**: Diode saturation current[14].
- **R_s / R_{sh}**: Series and Shunt resistances[15].
- **n**: Ideality factor[16].

D. Fill Factor (FF)

This represents the ratio of maximum power to the theoretical power17171717.

$$FF = \frac{P_m}{V_{oc} \times I_{sc}}$$

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3. Array Configuration Scaling (Se-P Topology)

The experiment utilized a 12-panel **Series-Parallel (Se-P)** array in a 3×4 arrangement[181818]. To calculate the final array values, you must scale the single-panel results:

- **Array V_{oc}**: $V_{oc,panel} \times 3$ (Number of panels in series)[19].
- **Array I_{sc}**: $I_{sc,panel} \times 4$ (Number of strings in parallel)[20].
- **Array V_m**: $V_{m,panel} \times 3$[21].
- **Array I_m**: $I_{m,panel} \times 4$[22].
- **Array P_m**: $V_{m,array} \times I_{m,array}$[23].

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4. Experimental Positioning Details

The study highlights that the output power depends heavily on the **variance** (mounting height) and **positioning**[24242424].

- **Flat Position (FP)**: Panels placed horizontally at 180°[25].
- **Highest Performance Setup**: Achieving 156 W with a flat-position array at a **93 cm variance** using Se-P interconnection[26].
- **Footprint Optimization**: While FP yields more power, V-shape (VP) and Inverted V-shape (IVP) are strategic for reducing the land footprint in urban environments[272727272727272727].

$$T_{cell} = T_\infty + 0.078 H_g$$

$$FF = \frac{P_m}{V_{oc} \times I_{sc}}$$

$$I = I_{ph} - I_0 \left(e^{\frac{q(V+IR_s)}{nkT}} - 1 \right) - \left(\frac{V + IR_s}{R_{sh}} \right)$$

$$P_{pv} = P_{stc} \times DF_{pv} \times (H_g / H_{stc}) \times [1 + \beta_p (T_{cell} - T_{stc})]$$

$$I_m = I_{m,STC} \times \left(\frac{H_g}{H_{STC}} \right) \times [1 + \alpha_{Im} (T_{cell} - T_{STC})]$$

$$V_m = V_{m,STC} \times [1 + \beta_{Vm} (T_{cell} - T_{STC})]$$

$$: I_{sc} = I_{sc,STC} \times \left(\frac{H_g}{H_{STC}} \right)$$

$$: V_{oc} \approx V_{oc,STC} + \frac{nkT}{q} \ln \left(\frac{H_g}{H_{STC}} \right)$$

code

```
import csv
```

```
from datetime import datetime, timedelta
import math
import random

def generate_summer_pv_csv(
    start_date="2025-04-01",
    months=3,
    output_file="pv_summer_apr_may_jun.csv"
):
    # ==============================
    # SYSTEM CONSTANTS (from paper)
    # ==============================
    FF = 0.67
    Voc_stc = 24.8
    Isc_stc = 0.60
    T_stc = 25

    n_series = 4
    n_parallel = 3

    beta_voc = -0.0038     # −0.38 % / °C
    alpha_isc = 0.001      # +0.1 % / °C

    current_loss_factor = 0.82
    voltage_factor = 0.84
    vm_ratio = 0.80

    # Time slots: 08:30 – 18:30 (summer longer sun)
    time_slots = [
        (datetime.strptime("08:30", "%H:%M") + timedelta(minutes=30*i)).strftime("%H:%M")
        for i in range(21)
```

```
]

start_dt = datetime.strptime(start_date, "%Y-%m-%d")
end_dt = start_dt + timedelta(days=months * 30)

with open(output_file, mode="w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow([
        "Date", "Time", "Irradiation_Wm2", "AmbientTemp_C", "CellTemp_C",
        "Voc_V", "Isc_A", "Vm_V", "Im_A", "Pm_W", "FF"
    ])

    current_date = start_dt

    while current_date < end_dt:
        # Summer ambient temperature
        T_ambient = random.uniform(32, 42)

        # Higher summer peak irradiation
        daily_peak = random.randint(1050, 1150)

        for idx, time_str in enumerate(time_slots):
            # Summer irradiation curve (wider bell)
            angle = (idx / (len(time_slots) - 1)) * math.pi
            irradiation = max(
                250,
                daily_peak * math.sin(angle) + random.uniform(-40, 40)
            )

            # Cell temperature (summer realistic)
            T_cell = T_ambient + (0.025 * irradiation / 10)
```

```
delta_T = T_cell - T_stc

# -------------------------------
# ELECTRICAL CALCULATIONS
# -------------------------------
Isc_panel = (
    Isc_stc *
    (irradiation / 1000) *
    (1 + alpha_isc * delta_T) *
    current_loss_factor
)
Isc_array = Isc_panel * n_parallel

Voc_panel = (
    Voc_stc *
    voltage_factor *
    (1 + beta_voc * delta_T)
)
Voc_array = Voc_panel * n_series

Vm_array = vm_ratio * Voc_array
Pm_array = FF * Voc_array * Isc_array
Im_array = Pm_array / Vm_array if Vm_array > 0 else 0

writer.writerow([
    current_date.strftime("%Y-%m-%d"),
    time_str,
    int(irradiation),
    round(T_ambient, 1),
    round(T_cell, 1),
    round(Voc_array, 2),
```

```
            round(Isc_array, 2),
            round(Vm_array, 2),
            round(Im_array, 2),
            round(Pm_array, 0),
            FF
          ])

        current_date += timedelta(days=1)

    print(f"✅ Summer CSV generated: {output_file}")

# RUN
generate_summer_pv_csv()
```